EN ENGLISH

Kibble Dispenser
Outdoor

User manual

Installation, use and maintenance



0.6.3 draft

design v6.3

⚠ **Preliminary: not yet printed or tested**

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Safety instructions

Read this manual before you install and use the unit, and keep it. If you pass the unit on, pass the manual on too.

Symbols in this manual

- Warning: risk of injury or damage. Read it before you go on.
- Check: make sure of this before the next step.
- Note: useful information.
- (A) Part letter, as listed in What's in the box.
- [1] Screw number, as listed in the hardware bag.

Intended use

- Kibble Dispenser Outdoor gives out portions of dry pet food from a food container that stands on a shelf above it, one portion at a time, at the press of a button.
- Use it only for dry pet food (kibble) with pieces up to 25 mm. It is not made for wet food, liquids or anything else.
- Install it under a shelf or a table top, indoors, or outdoors under a roof.
- It does not replace your care: check every day that your pet eats and that food comes out.
- This is a preliminary open-source design: no unit has been printed or tested yet. Test it for several days under supervision before you rely on it.

Electricity

- The unit runs on safe extra-low voltage (12 V DC). Use only the power supply from the kit.
- The power supply is not waterproof: put it in a dry place, protected from rain. Keep the mains plug and the socket dry, and lead the mains cable with a drip loop.
- Do not use the unit if a cable, the power supply or the housing is damaged.
- Unplug the power supply before you take the unit down, open it or clean it.

Moving parts

- Never reach into the outlet, the chute or the shelf hole while the unit is plugged in: the dosing disc turns when the button is pressed, and it can move briefly to its rest position when the power comes on.
- Switch the unit off and unplug it before you clear a blockage.

The gate

- Always turn the orange ring to CLOSED before you take the unit down and whenever the food container is removed: otherwise food falls out of the shelf hole.
- Keep a circle of 20 cm free around the unit so that you can reach the ring with your hand.

Children and pets

- This is not a toy. Do not let children play with the unit, its button or the packaging.
- Lead the cable and place the power supply where pets cannot chew them.
- Do not let pets climb on the unit, hang from it or play with it.

Outdoors

- Outdoors, install the unit only under a roof, out of driving rain and away from sprinklers. Never spray it with a hose or a pressure washer.
- Avoid full sun all day: heat spoils the food faster.
- Seal the joint between the food container and the shelf so that no rain gets into the food.

Food hygiene

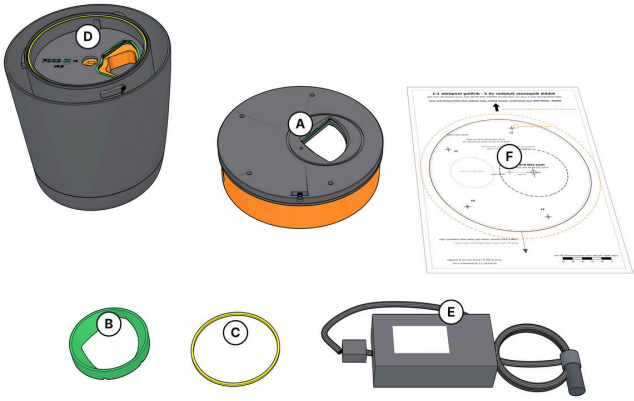
- Use only fresh dry pet food in good condition. Throw away food that is damp, mouldy or smells bad.
- Clean the food path every few months and whenever you change the food (page 24).
- The printed parts have fine grooves where crumbs can collect, and they are not certified as food-contact materials.
- Wash the parts by hand only, at 40 °C at most. No dishwasher.

During installation

- Wear safety glasses when you drill.
- Check that the shelf can carry the full container: 200 litres of dry food weigh about 80 kg.
- Make sure there are no cables or pipes where you drill.
- Use only the 12 V power supply from the kit. Keep the mains plug and the socket dry.
- Keep small parts away from children and pets.

What's in the box

The dispenser unit and the flange set come ready-made: do not take them apart. Check that every part is there.



(A) Flange set ×1
ready-made: two black plates pressed together, with the green gate blade, the orange ring and the soft yellow skirt

(B) Funnel collar ×1
green ring with 2 small pins

(C) Flange gasket ×1
soft yellow ring

(D) Dispenser unit ×1
ready-made and closed: the whole pod with the dosing disc, the electronics, the control panel and a short cable with plug

E **Power supply set** ×1
12 V power supply with screw-terminal adapter, supply cable with waterproof connector and yellow grommet

F **Drilling template** ×1
paper, 1:1, A4

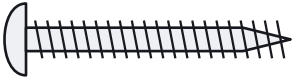
ⓘ You also need a food container (for example a drum) and a level shelf or table top. They are not in the kit.

ⓘ The drive inside is greased for life: you never need grease.

Hardware bag

Shown at actual size: hold a screw against the drawing to identify it.

1 **Wood screw 5 × 35** ×4
flange to shelf (5 × 30 for an 18 mm shelf)



2 **Cable tie 3.6 × 140, UV-resistant** ×1
strain relief for the cable in the flange

Tools you need

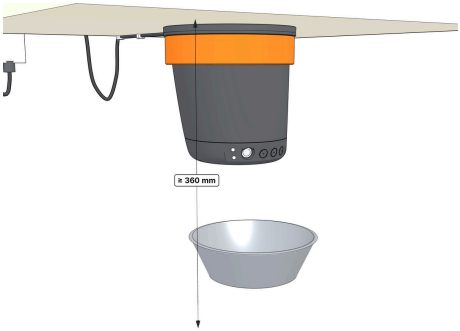
- Drill
- Wood drill bit Ø 3 mm
- Pencil and awl
- Side cutter
- Spirit level
- Hole saw Ø 76 mm
- Screwdriver PZ2
- Adhesive tape
- Cable clips for Ø 5 mm cable (not included)
- Safety glasses

Installation

PREPARE THE SHELF

1 Choose the place

Spirit level

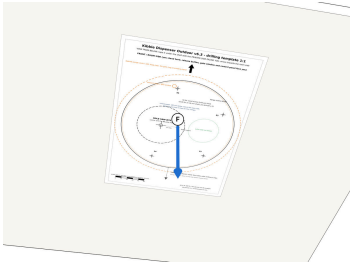


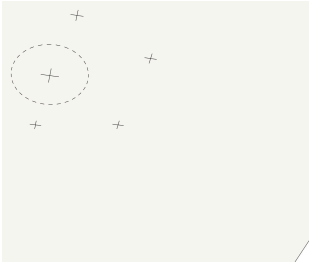
1. Pick a level shelf or table top, 18 to 40 mm thick. Outdoors, choose a place under a roof.
2. Leave at least 36 cm free below the shelf: 18.2 cm for the unit, 8 cm to take it down, and room for the pet bowl.
3. Keep a circle of 20 cm free around the unit: you turn the orange ring by hand.
4. The release button and the control panel face the room: keep 15 cm free in front of the panel for a finger. The cable leaves the flange at the back.

Check: The shelf is level in both directions.

2 Mark the holes with the template

 Ⓕ ×1 Pencil and awl Adhesive tape





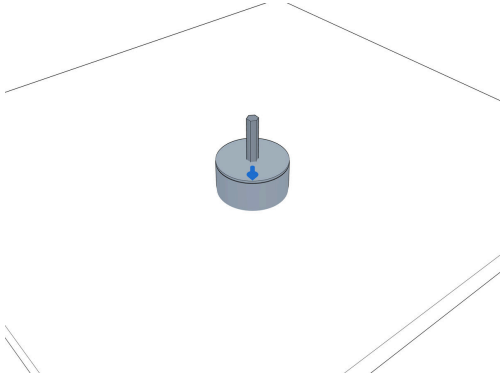
1. Print the drilling template Ⓕ at 100 % and check the 50 mm scale bar.
2. Tape it to the underside of the shelf with the printed side facing you and FRONT towards the room.
3. Prick through the paper with an awl: the centre of the big hole and the 4 screw points.
4. Take the template off.

Check: You have the centre of the big hole and 4 screw points.

The template is a separate A4 sheet: kibble-dispenser-drilling-template.pdf. Print it at 100 %, PRINTED SIDE FACING YOU under the shelf.

3 Drill the holes

Drill · Hole saw Ø 76 mm · Wood drill bit Ø 3 mm · Safety glasses



⚠ Wear safety glasses and hold the drill with both hands: the hole saw can jerk.

1. Drill straight through at the centre mark with the centre drill of the hole saw.
2. Saw the Ø 76 mm hole from above, guided by the pilot hole. Keep the drill upright.
3. Pre-drill the 4 screw points with the Ø 3 mm bit, about 20 mm deep.
4. Remove the chips and smooth the edges.

✓ **Check:** The big hole goes straight through the shelf.

4 Fit the flange set


Ⓐ ×1


Ⓑ ×1


Ⓒ ×1

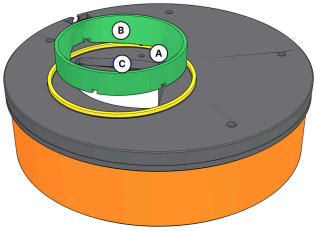

Ⓔ ×1

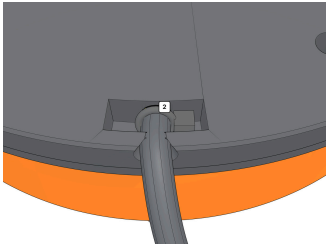
2 ×1


1 ×4


Side cutter


Screwdriver PZ2



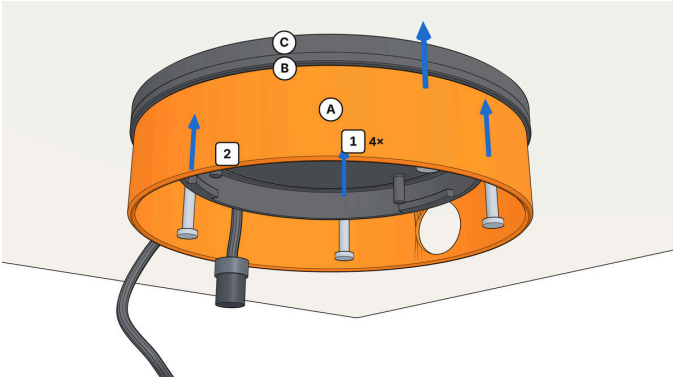


⚠ Shelf thinner than 20 mm: use 4 screws 5 × 30 instead, so the tips do not come through.

1. Press the flange gasket Ⓒ into the groove on the top of the flange set Ⓐ, and put the green collar Ⓑ into the big opening: its 2 pins go into the 2 small holes.
2. Push the bare end of the supply cable from the power supply set Ⓔ into the small hole inside the flange until it comes out at the back of the rim. Press the yellow grommet into the hole: the connector hangs below.
3. Pull the cable snug from outside. In the small window on the top of the flange, close the cable tie 2 tightly round the cable and cut off the tail.

4 Fit the flange set

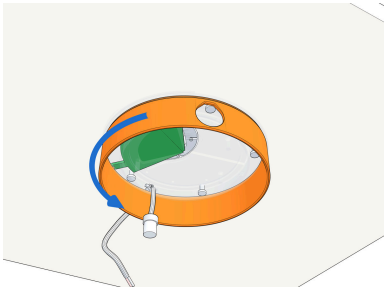
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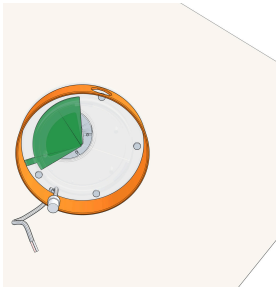


4. Lift the flange against the shelf, FRONT mark towards the room, cable at the back: the collar goes into the big hole.
5. Fix it with 4 wood screws [1] and the PZ2 screwdriver. Tighten until snug.

Check: The flange sits tight against the shelf. The collar does not stick out above the shelf.

5 Get to know the gate



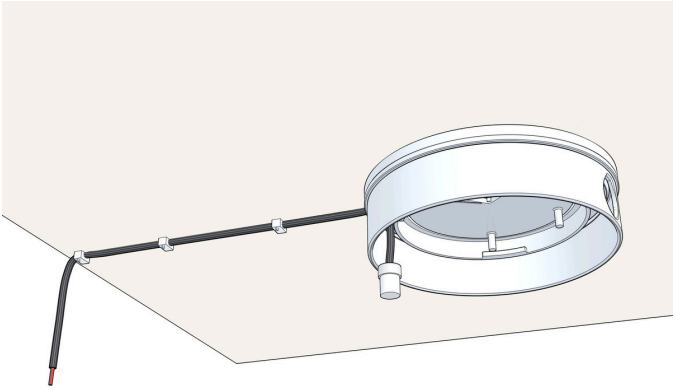


1. Turn the orange ring anticlockwise, seen from below, until it stops: the green blade closes the opening. This is CLOSED.
2. In CLOSED the round window of the ring sits where the release button of the unit will be.
3. Turn the ring back until it stops: the opening is free. This is OPEN.
4. Leave the ring in CLOSED for now.

Check: The ring turns smoothly between its two stops (about 1/3 of a turn).

6 Route the cable

Cable clips for Ø 5 mm cable (not included)



1. Lead the cable from the back of the flange along the underside of the shelf to the back edge.
2. Fix it with ordinary cable clips for Ø 5 mm cable, about every 20 to 30 cm. Do not pinch the cable.
3. Let the end hang down at the edge: the power supply comes in step 9.

ⓘ The cable tie inside the flange takes any pull on the cable: nothing reaches the connector.

HANG AND CONNECT

7 Connect the unit

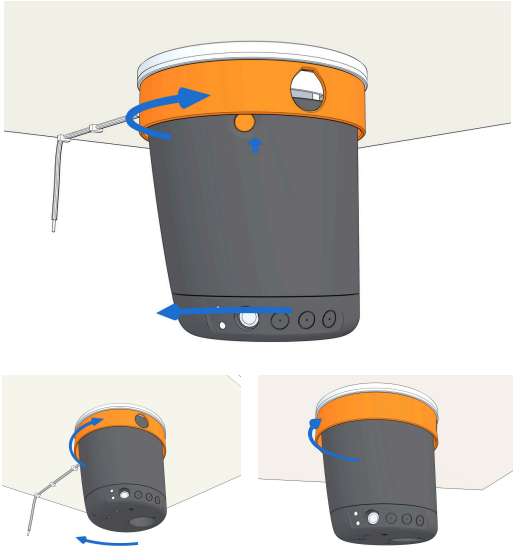
 Ⓓ ×1



1. Hold the dispenser unit Ⓓ about 8 cm below the flange, release button towards the room. A box or a stool under it helps.
2. Push the plug of the unit into the connector hanging from the flange and screw the coupling ring tight.
3. Push the connector and the loose cable down into the top of the shell.

Check: The coupling ring is screwed fully home.

8 Hang the unit

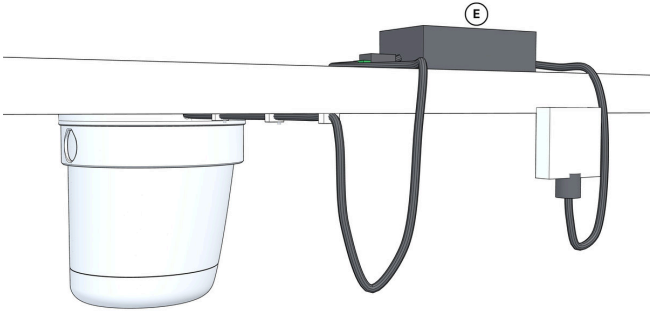


1. Check that the orange ring is in CLOSED.
2. Lift the unit straight up into the flange. Turn it a little until the 3 lugs slip into the slots and it rises.
3. Push up and turn it clockwise, seen from below, by about 1/12 of a turn (30°), until the latch clicks.
4. Turn the orange ring clockwise until it stops: OPEN. Food can now reach the unit.

Check: The unit does not turn back, and the gap under the flange is even all round.

9 Connect the power

 Ⓔ ×1 Screwdriver PZ2

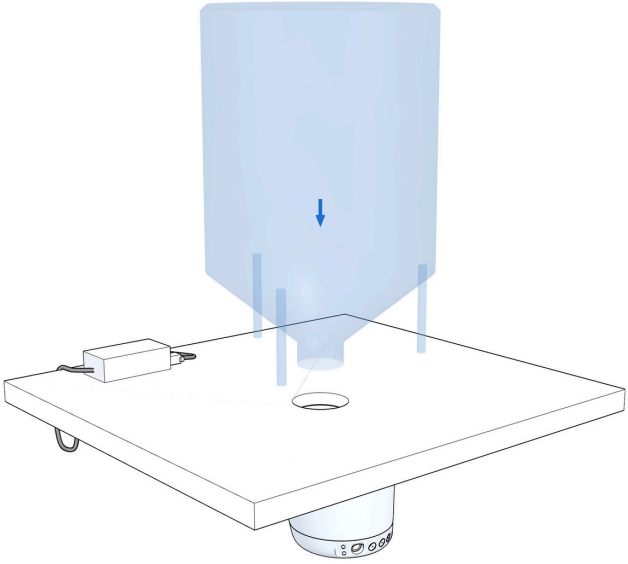


⚠ Mains voltage: the power supply, the plug and the socket must stay dry.

1. Put the power supply Ⓔ in a dry place, protected from rain, for example on the shelf under the roof.
2. Let the cable hang from the shelf edge in a loop that goes lower than the power supply (drip loop): water drips off at the bottom of the loop.
3. Connect the 2 wires to the adapter: red to +, black to −. Tighten the screws.
4. Plug the power supply into the socket.

✓ **Check:** The unit starts switched OFF, with the status LED off. It never gives food when the power comes on.

10 Place the container and fill it



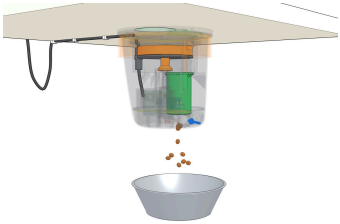
⚠ A full container is heavy: make sure the shelf can carry it.

1. Put your food container on the shelf, with its bottom opening right over the green collar.
2. Seal the joint so that no rain gets in.
3. Fill it with dry food, pieces up to 25 mm.

✓ **Check:** With the ring OPEN, food reaches the top of the unit.

First use

11 First test



1. Put the pet bowl under the outlet.
2. The control panel is on the room side of the cap, under the release button. Press its button twice quickly (double press): the status LED lights, a sound plays and the unit gives one portion.
3. Now press the button once: after a short moment the unit gives one portion with a sound. Repeat a few times.
4. The first portions can be smaller until the disc fills up.

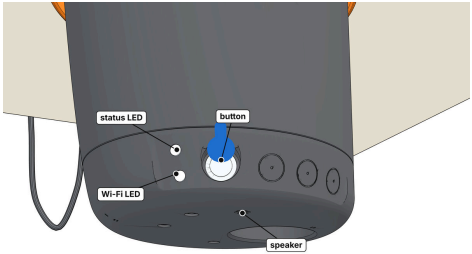
Check: Food falls into the bowl at every single press.

THE FIRST DAYS

- Check every day that the food reaches the bowl and that your pet eats it.
- Weigh 10 portions on a kitchen scale: then you know how much one press gives with your food.
- Some pets need a few days to get used to the sound.

Everyday use

Everyday use: one button



- The control panel has one button and two lights. Press the button with a fingertip, right to the bottom of the recess.
- Double press (twice within 0.4 s): switches the unit ON or OFF. Switching ON gives one portion; switching OFF gives none.
- Single press: one portion, only while the unit is ON. While it is OFF the status LED only blinks. Holding the button never gives more.
- Hold 5 s: resets all settings except Wi-Fi. Hold 10 s: full reset, Wi-Fi included. Let go after the LED signal; no food is given.
- Top LED: status (ON/OFF). Bottom LED: Wi-Fi. A sound plays at every portion; one portion is about 6.5 g.

ⓘ The unit never gives food when the power comes on: after a power cut it starts OFF. The button behaviour and the meaning of the LEDs are provisional until the firmware is updated (GitHub issue #2).

Using the gate



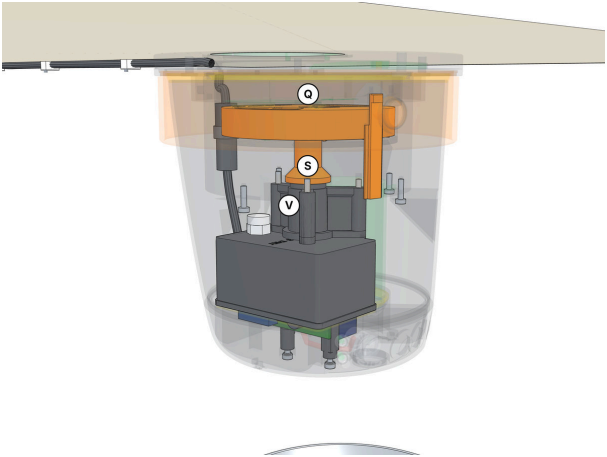
⚠ With the unit taken down, keep the gate CLOSED: otherwise food falls out of the shelf hole.

- OPEN (ring turned clockwise, seen from below): food flows into the unit. This is the normal position.
- CLOSED (ring turned anticlockwise to the stop): the green blade holds the food in the container.
- Close the gate before you take the unit down. Only then does the window of the ring free the release button.
- You can refill the container with the gate open or closed.

⚠ Always turn the ring to CLOSED before you take the unit down or remove the food container.

Maintenance

Almost no maintenance



- No grease and no oil, ever: the drive inside is sealed and greased for life.
- No silica gel to change: the electronics box breathes through a membrane vent.
- The dosing disc cleans itself: small ribs underneath sweep crumbs away at every turn.
- Keep the tiny holes of the two drains in the flat bottom free of dirt; rinse them with a little water if needed.
- For hygiene, clean the food path every few months and whenever you change the food (next two steps).

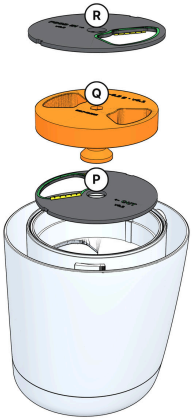
12 Take the unit down



1. Switch the unit off and unplug the power supply.
2. Turn the orange ring anticlockwise, seen from below, to the stop: CLOSED. The window now shows the release button.
3. Press the button through the window and turn the unit anti-clockwise by 30°.
4. Lower it about 8 cm, unscrew the connector in the gap and take the unit down in one piece.

ⓘ The ring can only free the button when the gate is closed, so no food falls out.

13 Clean the food path



⚠ No dishwasher, no oil, no grease.

1. Stand the unit on the table. Lift out the upper plate, the disc and the lower plate.
2. Wash them and the inside of the housing by hand with warm water (max. 40 °C) and a little dish soap. Leave the lips in place.
3. Let everything dry completely.
4. Refit the lower plate, the disc and the upper plate, each turned until it drops into place. Then connect and hang the unit again (steps 7 and 8).

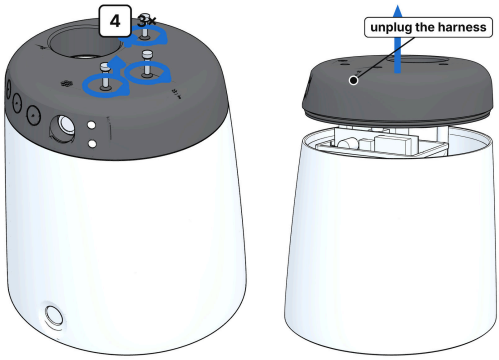
✓ **Check:** Everything is dry before you refit it.

Service only: not part of normal use or maintenance

14 Open the cap (service only)

Torx T10 screwdriver, long blade

Soft cloth



Unplug the power supply first. Do not pull the cap off in one go: you would tear the panel cable.

1. Only for repairs or if water got in. Take the unit down (step 12) and stand it upside down on a soft cloth.
2. Undo the 3 screws in the flat bottom with a T10 screwdriver. Keep the screws and their small O-rings.
3. Lift the cap only a little: the control panel is wired to the board. Unplug the panel harness, then take the cap off.
4. To close: plug the harness back in, put the cap on and tighten the 3 screws with their O-rings by hand only.

Troubleshooting

PROBLEM CAUSE · REMEDY

1 No food comes out

CAUSE	The gate is closed, the unit is OFF (status LED off), the container is empty, or food is stuck above the collar.
REMEDY	Turn the ring to OPEN, double-press the button, refill, tap the container.

2 The button does nothing

CAUSE	No power, the press was too light, or the unit is OFF: a single press then only makes the status LED blink.
REMEDY	Check the power supply. Press with a fingertip right to the bottom of the recess; double-press to switch ON.

3 The disc stops, or you hear grinding

CAUSE	A piece bigger than 25 mm or a hard piece is stuck at an edge.
REMEDY	Switch off, take the unit down (step 12) and clean it (step 13).

4 Portions too small or too big

CAUSE	The portion depends on the food: about 5.4 to 7.4 g from light to heavy kibble.
REMEDY	Press more or fewer times.

5 Water inside

CAUSE	A leaking container joint, a blocked drain or a damaged seal.
REMEDY	Seal the container joint and clean the two drains. Water in the electronics box: unplug, take the unit down, open the cap (step 14), let it dry and check the seals.

6 Noise

CAUSE	A soft whirr at each portion is normal. Squeaking or scraping: crumbs between the disc and a plate.
REMEDY	Clean the food path. Never oil anything.

7 The unit does not lock

CAUSE	The rim seal is out of its groove, or the lugs are not in the slots.
REMEDY	Seat the rim seal, find the slots, push up firmly while you turn.

8 The release button cannot be reached

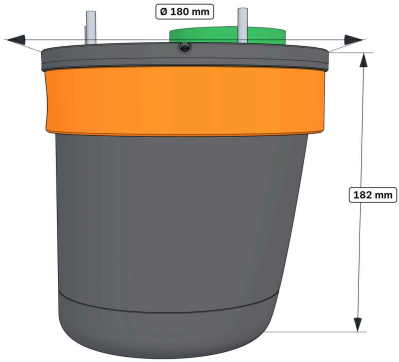
CAUSE	The gate is not fully closed: the ring covers the button.
REMEDY	Turn the ring anticlockwise, seen from below, to the stop.

9 Settings wrong or Wi-Fi lost

CAUSE	A setting was changed by mistake, or the network changed.
REMEDY	Hold the button 5 s to reset the settings (Wi-Fi is kept), or 10 s for a full reset.

Still not working? Unplug the unit and look for help on the project site (page 34).

Technical data



Design	Kibble Dispenser Outdoor v6.3
Size under the shelf	Ø 180 mm (flange), 182 mm high
Weight of the unit	about 1.3 kg
Shelf	level, 18–40 mm thick; one hole Ø 76 mm
Free space	at least 36 cm below the shelf, Ø 20 cm round the unit
Portion	about 6.5 g (standard disc); 5.4–7.4 g depending on the food
Food size	dry food up to 25 mm

Gate	in the flange, turned by the orange ring (113°); the unit can only be released with the gate closed
Controls	1 recessed IP67 push-button, 2 LEDs (status, Wi-Fi), speaker
Power	12 V DC, power supply included; M8 IP67 connector inside the pod; cable leaves the flange at the back, with a cable-tie strain relief
Weather protection	electronics box designed for IP66, inside a sealed outer pod with two insect-proof labyrinth drains (not tested yet)
Strength	calculated for 250 N down and 100 N sideways at the bottom (not tested yet)
Materials	PETG, TPU 95A seals, NBR O-rings, stainless steel screws and dowel pin, brass threads, UV-resistant PA66 cable tie
Maintenance	no grease, no silica gel; the cap opens only for service (3 screws T10)

All values are design values: nothing has been printed or tested yet.

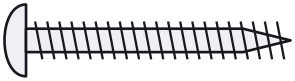
Spare parts and printable parts

Every printed part can be printed again from the open design files on the project site. Screws, O-rings, the electronics and the power supply are standard bought parts: the parts list in the design files gives their exact type.

A		**Flange set** Ready-made module: replace it as a whole	×1
B		**Funnel collar** PETG green	×1
C		**Flange gasket** TPU 95A yellow	×1
D		**Dispenser unit** Ready-made module: replace it as a whole	×1
E		**Power supply set** Bought part	×1
F		**Drilling template** PDF: print it at 100 %	×1

ⓘ The printable parts of the unit and the flange set, with their colours, are listed in the appendix (page 36).

SCREWS

1	**Wood screw 5 × 35** flange to shelf (5 × 30 for an 18 mm shelf)  1:1	×4
2	**Cable tie 3.6 × 140, UV-resistant** strain relief for the cable in the flange	×1

ⓘ Print the seals and lips in TPU 95A and all other printed parts in PETG, in the colours shown. The electronics box and its cap must be printed solid (100 % infill) to stay watertight; the print settings are in the design files.

Disposal



Crossed-out wheeled bin: do not put it in household waste.

The electronics box contains electronic parts. Do not put it in household waste: take it to a collection point for electrical and electronic equipment (WEEE). Dispose of the plastic parts and screws according to local rules. The unit contains no battery.

Printed parts: PETG and TPU. Many household plastic collections do not take them: ask your local recycling centre.

Open source and licences

Kibble Dispenser is open source: design files under CERN-OHL-P-2.0, software under the MIT licence, these instructions under CC BY 4.0. © 2026 Enrico Icardi and contributors.

Hardware (design files)	**CERN-OHL-P-2.0**
Software (firmware)	**MIT**
Documentation (this manual, pictures)	**CC BY 4.0**

https://ricricucit.github.io/kibble-dispenser/

Project site: 3D instructions, design files, drilling template and updates

Drilling template

The 1:1 drilling template is a separate A4 PDF (kibble-dispenser-drilling-template.pdf, also on the project site). Print it at 100 % (actual size, no scaling) and check the 50 mm scale bar with a ruler. Tape it under the shelf with the PRINTED SIDE FACING YOU and FRONT towards the room.

Open-source hardware is provided as is, without any warranty: see the licence texts.

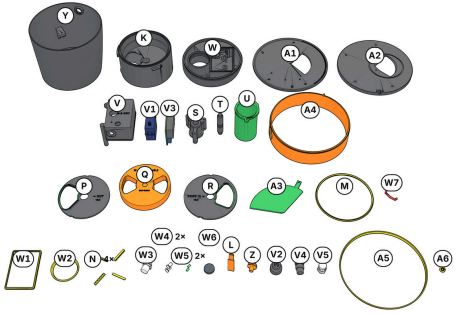
APPENDIX A

Build it yourself

If you print and build the unit and the flange set yourself, do these steps first, then install them like the kit. The online 3D instructions show every step as an animation that you can turn and zoom.

https://ricricucit.github.io/kibble-dispenser/
3D instructions online

Print the parts as described in the print notes of the design. You then build the unit and the flange set yourself.



Printed and bought parts

K		**Dosing housing** PETG black	×1

L		**Safety latch** PETG orange	×1
M		**Rim seal** TPU 95A yellow	×1
N		**Comb lips** TPU 95A yellow	×4
P		**Lower plate** PETG black, PETG green	×1
Q		**Dosing disc** PETG orange, PETG black	×1
R		**Upper plate** PETG black, PETG green	×1
S		**Tower** PETG black	×1
T		**Shaft** PETG black	×1
U		**Chute** PETG green	×1
V		**Electronics box** PETG black + bought parts	×1
V1		**Servo** Bought part	×1
V2		**Hub** PETG black + bought parts	×1
V3		**Circuit board** Bought part	×1
V4		**Cable gland M12** Bought part	×1
V5		**Membrane vent M12** Bought part	×1

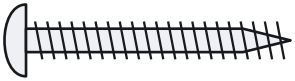
W		**Cap** PETG black	×1
W1		**Cap seal** TPU 95A yellow	×1
W2		**Chute lip ring** TPU 95A yellow	×1
W3		**Panel push-button** Bought part	×1
W4		**Light pipes** Bought part	×2
W5		**LEDs 3 mm** Bought part	×2
W6		**Speaker with membrane** Bought part	×1
W7		**Panel harness** Bought part	×1
Y		**Shell** PETG black	×1
Z		**Release button** PETG orange + bought parts	×1
A1		**Lower flange plate** PETG black	×1
A2		**Upper flange plate** PETG black	×1
A3		**Gate blade** PETG green	×1
A4		**Gate ring** PETG orange	×1
A5		**Skirt lip** TPU 95A yellow	×1



A6 **Cable grommet** ×1
TPU 95A yellow

Screws, O-rings and small parts

1 **Wood screw 5 × 35** ×4
flange to shelf (5 × 30 for an 18 mm shelf)

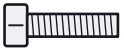


2 **Cable tie 3.6 × 140, UV-resistant** ×1
strain relief for the cable in the flange

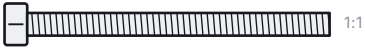
3 **Thread-forming screw PT K30 × 10, T10** ×2
chute



4 **Thread-forming screw PT K30 × 12, T10** ×6
shell (3) and cap (3)



5 **Screw M3 × 40** ×3
box + tower + housing



6 **O-ring 3 × 1** ×5
under the screw heads: 2 long screws, 3 cap screws

7 **Screw M3 × 8 (+ washer)** ×5
servo (4) and hub (1)



8 **Heat-set insert M3** ×7
housing (3) and box (4)

9	**O-ring 9 × 2** shaft	×2
10	**O-ring 24 × 2** tower to box	×1
11	**O-ring 132 × 2** cap seam	×1
12	**O-ring 14 × 1.5** release button	×1
13	**O-ring 5 × 1** light pipes	×2
14	**Dowel pin 4 × 6 (ISO 2338)** pivot of the gate blade	×1
15	**Food-grade silicone grease (NSF H1)** all O-rings, thinly	×1

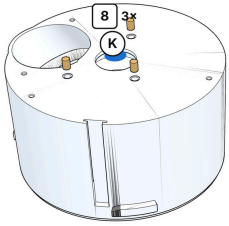
Extra tools

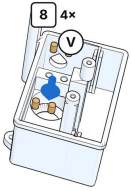
- Soldering iron with insert tip
- Hex key 2.5 mm, long
- Torx T10 screwdriver, long blade
- Soft cloth

ⓘ Grease every O-ring thinly with food-grade silicone grease before you fit it. The electronics are not final: wire the board as described in the electronics notes.

A1 Fit the threaded inserts

[8] ×7 Soldering iron with insert tip



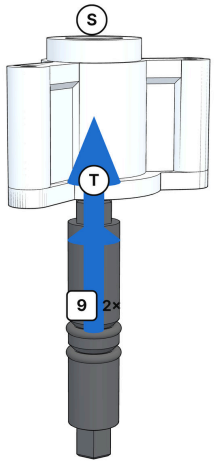


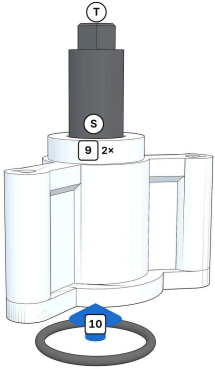
The iron tip is hot: work on a heat-proof surface.

1. Heat each insert [8] with the soldering iron and press it straight into its hole until it is flush.
2. Housing (K): 3 inserts in the bottom, for the long screws. Electronics box (V): 4 inserts in the servo cradle.
3. Let them cool before you screw into them. All other screws cut their own thread in the plastic.

Check: Every insert sits straight and flush with the surface.

A2 Build the drive module





T ×1 S ×1 9 ×2 10 ×1 15 ×1

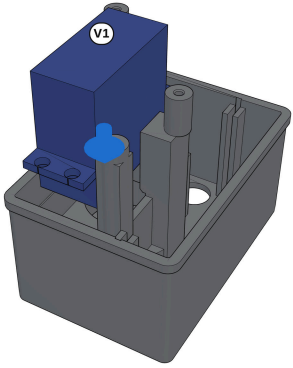
1. Grease the 2 O-rings 9 and roll them into the 2 grooves of the shaft T.
2. Push the shaft into the tower S from below, turning it slightly, until its hexagon sticks out at the top.
3. Grease the large O-ring 10 and press it into the groove under the tower.

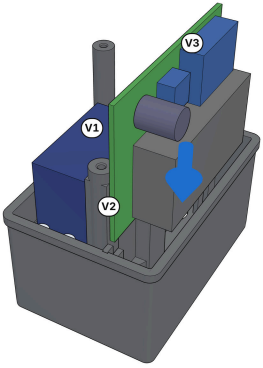
Check: The shaft turns by hand with a light, even drag.

A3 Fit out the electronics box

 V ×1  V1 ×1  V2 ×1  V3 ×1  V4 ×1

 V5 ×1  7 ×5  Hex key 2.5 mm, long



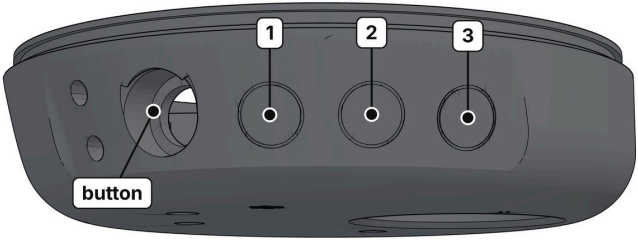


1. Set the servo (V1) to its zero position (electronics notes), put it in the cradle and fix it with 4 screws M3 × 8 and washers [7].
2. Push the hub (V2) onto the servo spline with its slot pointing to the gland side, and fix it with 1 screw M3 × 8 [7].
3. Slide the board (V3) up into its slots, parts facing outwards.
4. Screw the cable gland (V4) into the side wall and the membrane vent (V5) into the roof.

Check: The box is clean and dry inside; the short cable with the plug comes out of the gland.

A4 Optional: holes for extra buttons

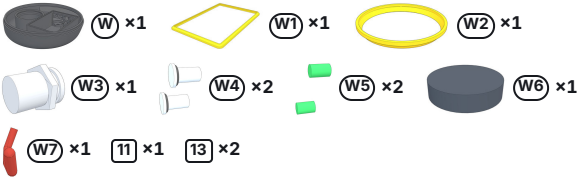
Drill · Wood drill bit Ø 3 mm · Step drill 4–12 mm · Safety glasses

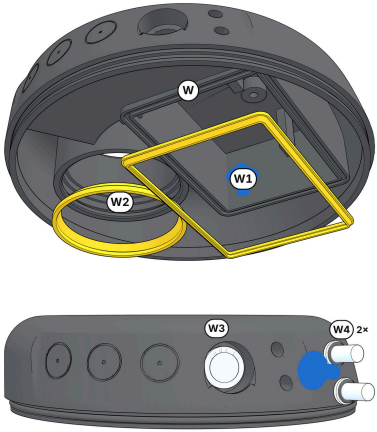


> ⚠ For experienced builders only: a badly drilled hole lets water into the electronics box. The firmware does not use extra buttons yet.

1. To the right of the button the cap has 3 flush knock-outs, each marked by an engraved ring and a centre dot. Leave them closed unless you add a button.
2. To open one, lay the cap on a soft pad. Drill a Ø 3 mm pilot at the centre dot, square to the face, then open it from outside with a step drill up to the Ø 12 step.
3. Deburr the inside edge and blow out the chips. Fit a 12 mm IP67 switch (nut up to 14 mm across flats) with a thin bead of neutral-cure silicone under its gasket.

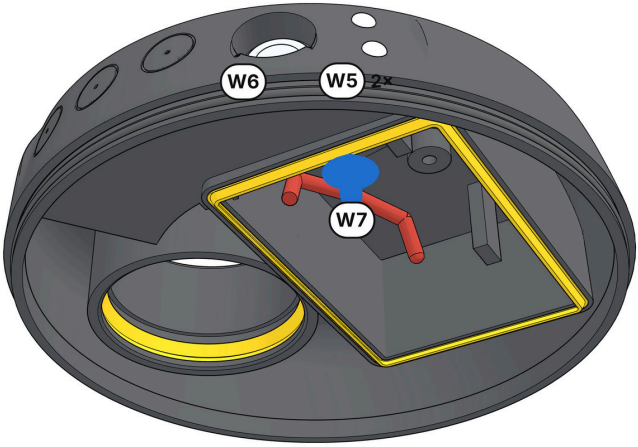
A5 Prepare the cap and the control panel





1. Press the cap seal (W1) into its D-shaped groove and the chute lip ring (W2) into the round socket. Grease the large O-ring [11] and roll it into the groove round the rim.
2. From outside: push the panel button (W3) into the recessed hole and tighten its nut from inside. Push the 2 light pipes (W4) with their O-rings [13] into their holes.

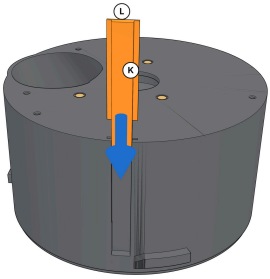
A5 Prepare the cap and the control panel
CONTINUED



3. From inside: put the 2 LEDs (W5) behind the light pipes (status on top) and the speaker with its membrane (W6) over the grille in the bottom.
4. Connect the panel parts to the harness (W7). Its plug goes onto the board when you close the box.

Check: The button clicks and springs back. The harness plug hangs free.

A6 Turn the housing over and fit the latch

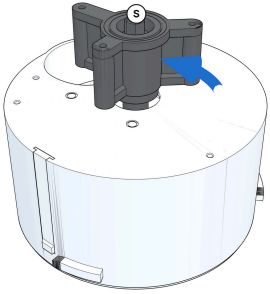


Ⓚ ×1 Ⓛ ×1

1. Lay a soft cloth on the table. Put the dosing housing Ⓚ on it upside down.
2. Slide the safety latch Ⓛ into its slot in the bottom, small bump outwards, until it is flush.

✓ **Check:** The latch does not stick out of the bottom.

A7 Set the drive module on the housing



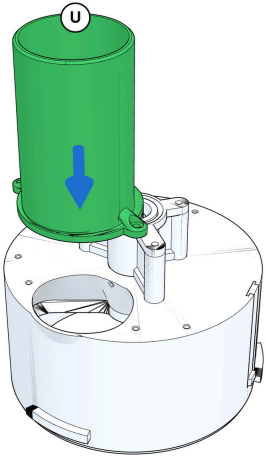
Ⓢ ×1

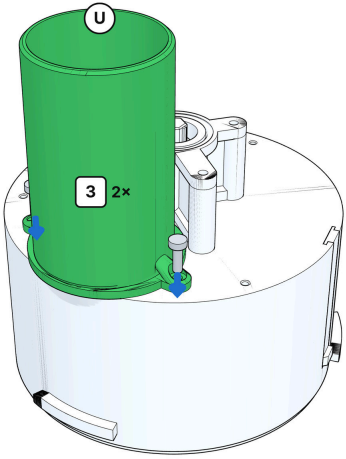
1. Set the drive module on the centre of the housing: the collar goes into the hole, the shaft into the housing.
2. Turn it until its 3 pillars sit over the 3 inserts. It fits only one way. No screws yet.

✓ **Check:** The module sits flat in the centre.

A8 Fit the chute

 Ⓤ ×1  3 ×2 Torx T10 screwdriver, long blade





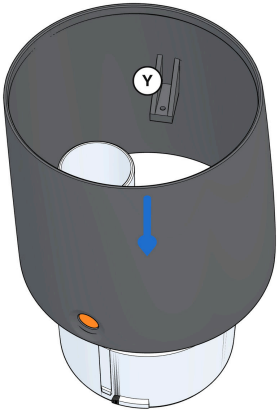
1. Put the green chute Ⓤ on the funnel outlet next to the drive module. It fits only one way.
2. Fix it with 2 screws PT K30 × 10 [3] and the long T10 screwdriver. They cut their own thread: tighten by hand, without force.

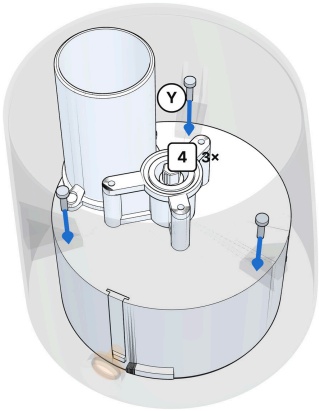
✓ **Check:** The chute stands straight and sits tight against the housing.

A9 Fit the shell

 Ⓨ ×1  Ⓩ ×1 12 ×1  4 ×3

 Torx T10 screwdriver, long blade





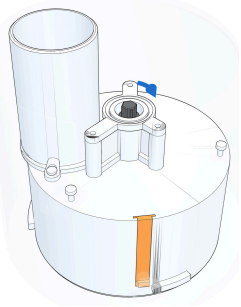
1. Grease the small O-ring 12, fit it in the groove of the button hole, and push the release button Ⓩ in from the inside.
2. Lower the shell Ⓨ over the chute and the drive module, with the orange button right next to the orange latch.
3. Fix it with 3 screws PT K30 × 12 4 through its 3 tabs, reaching in with the long T10 screwdriver. Tighten by hand.

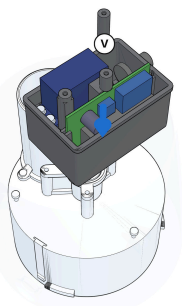
Check: The shell sits evenly all round.

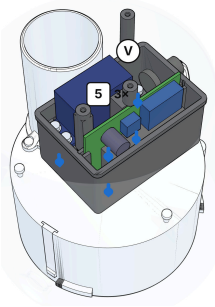
A10 Fit the electronics box

 Ⓥ ×1  5 ×3 6 ×2

 Hex key 2.5 mm, long





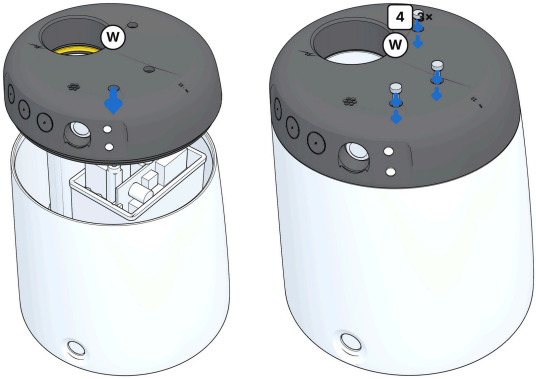


1. Turn the end of the shaft until its small rib points to the orange latch.
2. Lower the box Ⓥ, open side up, into the shell onto the drive module: the shaft goes into the hub. It fits only one way.
3. Put an O-ring 6 under the heads of the 2 screws that sit inside the box.
4. Screw in the 3 long screws M3 × 40 5 with the long hex key. Tighten until snug.
5. Lay the short cable along the housing, so that its plug sits just under the rim of the shell.

Check: The box sits flat on the drive module, without a gap.

A11 Close the box with the cap

 Ⓦ ×1  4 ×3 6 ×3  Torx T10 screwdriver, long blade

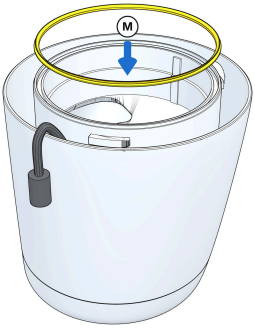


⚠ The screws cut their thread in the plastic: tighten by hand only (about 0.6 N·m). Open the cap only for service, at most about 5 times.

1. Plug the panel harness (W7) into the board.
2. Put the cap (W) on the box and the shell: the chute goes into the round socket of the cap. It fits only one way.
3. Put an O-ring [6] under the head of each of the 3 screws PT K30 × 12 [4]. Screw them in through the flat bottom with the T10 screwdriver: they pull the cap home.

✓ **Check:** The seam is closed all round. The 3 screw heads sit flush in their recesses.

A12 Stand the unit up and fit the rim seal



 Ⓜ ×1

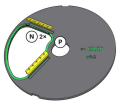
1. Turn the unit over: it now stands on its flat bottom. Pull the plug out at the top of the shell and let it hang outside.
2. Press the rim seal Ⓜ into the groove on the top rim of the housing, all the way round. The thin lip points outwards and up.

✓ **Check:** The seal sits evenly in the groove and the lip is not folded.

A13 Press the lips into the plates

 Ⓝ ×4  Ⓟ ×1

 Ⓡ ×1

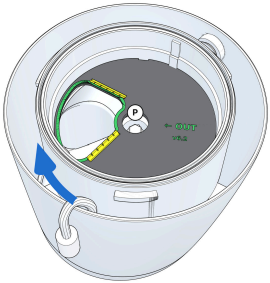




1. The 4 comb lips Ⓝ are all the same. Each fits either way round.
2. Lower plate Ⓟ, marked OUT: press 2 lips into the 2 grooves on its top, along the opening.
3. Upper plate Ⓡ, marked FOOD IN: turn it over and press the other 2 lips into the grooves on its underside.
4. The slanted soft flap of each lip leans towards the opening.

✓ **Check:** The lips sit fully in their grooves and stand up evenly.

A14 Fit the lower plate

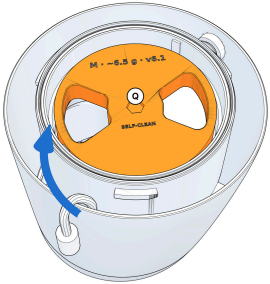


 Ⓟ ×1

1. Lower the lower plate Ⓟ into the housing, lips up, with the green OUT area over the chute.
2. Turn it until its 2 notches slide down onto the 2 ribs. It fits only one way.

✓ **Check:** The plate lies flat on its step.

A15 Fit the dosing disc



 Ⓠ ×1

1. Put the dosing disc Ⓠ on the shaft, label up.
2. Turn it slowly until it drops down. It fits only one way.

✓ **Check:** The disc sits level, with the label on top.

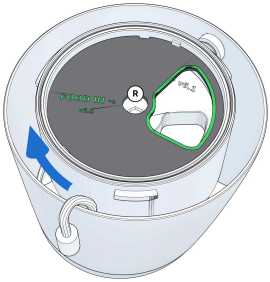
A16 Fit the upper plate



Ⓡ ×1

1. Lower the upper plate Ⓡ onto the disc, lips down, FOOD IN on top.
2. Turn it until its 2 notches slide down onto the 2 ribs. It fits only one way.

Check: The plate lies flat on its step, just below the rim. The unit is complete.



A17 Build the flange set

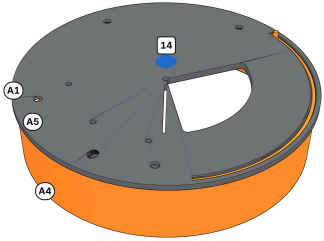

A1 ×1

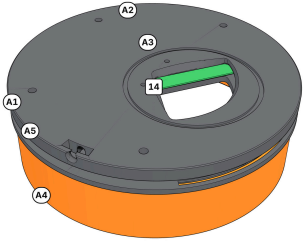

A2 ×1


A3 ×1


A4 ×1


A5 ×1 14 ×1





1. Hold the lower flange plate A1 with its 3 hooks pointing down. Press the skirt lip A5 into the groove underneath and snap the orange ring A4 on from below.
2. Press the dowel pin 14 into the centre hole and put the gate blade A3 on it, with the hole of its arm over the peg of the ring.
3. Put the upper plate A2 on and press it down: its 3 pins go into the lower plate and hold the set together.
4. The grommet A6 goes onto the supply cable: you fit it with the cable when you mount the flange (step 4).

Check: Turning the ring moves the blade between OPEN and CLOSED.

The unit and the flange set are ready. Now install them like the kit: continue with step 1, page 10.

APPENDIX B

Leak test at home

Optional, for makers: a screening of the seals, not a certification.

1. Take the electronics out of the box, or test before you fit them. Never test with the power connected.
2. Put strips of dry paper towel in the box (at the cable gland, the vent, the 2 screw heads in the roof, along the cap seal) and in the cap behind the control panel (button, light pipes, speaker), plus a few inside the shell, one of them next to the cable grommet under the flange.
3. Close the cap with its 3 screws and O-rings, and hang the unit in its flange set under a shelf or board, as in use, with the gate OPEN.
4. Spray it with a garden hose and a jet nozzle (about 6 mm, about 12.5 L/min: a 10 L bucket fills in 48 s) from 2.5–3 m, from every side and from below, for 3–4 minutes. Spray the control panel too.
5. Dry the outside, take the unit down and open the cap over a towel, lifting it only a little because of the panel cable.

Check: Dry strips in the box: passed. A wet strip shows where water gets in.

For a finer check, weigh a 10 g silica-gel bag on a 0.01 g scale and put it in the box: if it gains less than 0.05 g, no water got in.

Notes

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Kibble Dispenser Outdoor

User manual · English

https://ricricucit.github.io/kibble-dispenser/

Scan for the 3D instructions and the design files

0.6.3 draft · design v6.3
Preliminary: not yet printed or tested
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